

SAP ENGINEERING WORKS

Manufacturers & Exporters of High-Precision Industrial Flanges, Pipe Fittings & Fasteners

Product Range: Stainless Steel, Carbon Steel & Alloy Steel Industrial Components
Standards: ASME / ANSI / BS / IS / DIN / EN / JIS

1. MASTER PRODUCT LINE OVERVIEW

MATERIAL CATEGORY	GRADES & SPECIFICATIONS	TYPE / ITEMS OFFERED	DIMENSIONAL STANDARDS
Carbon Steel Flanges	ASME SA105, ASME SA516	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10 (Table D,E,F,H,J,K,R,S,T), IS 6392, JIS B220, AWWA, EN 1092-1 (PN-10,16,25,40,63,100)
Mild Steel Flanges	IS 2062	Slip On Flanges, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Stainless Steel Flanges	ASME SA182 304/304L/304H, 316/316L/316H, 321/321H, ASME SA240 304, 316, 321	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Alloy Steel Flanges	ASME SA182 F91, F92, F11 (CI 1,2,3), F22 (CI 1,3)	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Carbon Steel Fittings (Buttweld)	ASME SA234 WPB	90°/45° LR & SR Elbows, 90°/45° 3D Elbows, Equal/Unequal Tees, Concentric/Eccentric Reducers, Equal Cross, 180° LR/SR U Bends, End Caps	ASME B16.9
Stainless Steel Fittings (Buttweld)	ASME SA403 304 (H,L,N,LN), 316 (H,L,N,LN), 321 (H)	90°/45° LR & SR Elbows, Tees, Reducers, Crosses, 180° U Bends, End Caps	ASME B16.9
Alloy Steel Fittings (Buttweld)	ASME SA234 WP1, WP12, WP11, WP22, WP5, WP9, WP91, WP911 (CI 1,2,3)	90°/45° LR & SR Elbows, 3D Elbows, Reducers, Tees, Crosses, 180° Bends, End Caps	ASME B16.9
Forged Fittings (Carbon Steel)	ASME SA105	90°/45° Elbows, Concentric Reducers, Equal/Unequal Tees, Full/Half Couplings (SW / Threaded)	ASME B16.11
Forged Fittings (Stainless Steel)	ASME SA182 304, 316, 321 (H, L, N, LN)	90°/45° Elbows, Reducers, Tees, Couplings (Socket Weld / Threaded)	ASME B16.11
Forged Fittings (Alloy Steel)	ASME SA182 F91, F92, F11 (CI 1,2,3), F22 (CI 1,3)	90°/45° Elbows, Reducers, Tees, Couplings (Socket Weld / Threaded)	ASME B16.11

2. FLANGE DIMENSIONS (ASME B16.5)

ASME B16.5 - Class 150 Flanges (Dimensions in mm)

NPS	DN	O (OD)	FLANGE THK (TF)		HUB DIA (X)	BORE (B)			LENGTH THRU HUB (Y)			RAISED FACE		DRILLING	
			SO/SW/THRD	WN		SO/SW	WN	SO/SW/THRD	LJ	WN	DIA (R)	THK (TR)	PCD (W)	HOLE (D)	NO (N)
1/2"	15	90.0	9.6	11.2	30.0	22.2	22.9	16.0	16.0	48.0	34.9	2	60.3	15.9	4
3/4"	20	100.0	11.2	12.7	38.0	27.7	28.2	16.0	16.0	53.0	42.9	2	69.9	15.9	4
1"	25	110.0	12.7	14.3	49.0	34.5	34.9	18.0	17.0	56.0	50.8	2	79.4	15.9	4
1 1/4"	32	115.0	14.3	15.9	59.0	43.2	43.7	21.0	21.0	58.0	63.5	2	88.9	15.9	4
1 1/2"	40	125.0	15.9	17.5	65.0	49.5	50.0	23.0	22.0	62.0	73.0	2	98.4	15.9	4
2"	50	150.0	17.5	19.1	78.0	61.9	62.5	26.0	25.0	64.0	92.1	2	120.7	19.0	4

NPS	DN	O (OD)	FLANGE THK (TF)		HUB DIA (X)	BORE (B)			LENGTH THRU HUB (Y)			RAISED FACE		DRILLING	
			SO/SW/ THRD	WN		SO/ SW	WN	SO/SW/ THRD	LJ	WN	DIA (R)	THK (TR)	PCD (W)	HOLE (D)	NO (N)
2 1/2"	65	180.0	20.7	22.3	90.0	74.6	75.4	29.0	29.0	70.0	104.8	2	139.7	19.0	4
3"	80	190.0	22.3	23.9	108.0	90.7	91.4	31.0	30.0	70.0	127.0	2	152.4	19.0	4
3 1/2"	90	215.0	22.3	23.9	122.0	103.4	104.1	32.0	32.0	72.0	139.7	2	177.8	19.0	8
4"	100	230.0	22.3	23.9	135.0	114.3	116.8	34.0	33.0	77.0	157.2	2	190.5	19.0	8
5"	125	255.0	22.3	23.9	164.0	143.8	144.4	37.0	36.0	89.0	185.7	2	215.9	22.2	8
6"	150	280.0	23.9	25.4	192.0	170.7	171.4	40.0	40.0	89.0	215.9	2	241.3	22.2	8
8"	200	345.0	27.0	28.6	246.0	221.5	222.2	45.0	44.0	102.0	269.9	2	298.5	22.2	8
10"	250	405.0	28.6	30.2	305.0	275.2	277.4	50.0	49.0	102.0	323.8	2	362.0	25.4	12
12"	300	485.0	30.2	31.8	365.0	327.0	328.2	56.0	56.0	115.0	381.0	2	431.8	25.4	12
14"	350	535.0	33.4	35.0	400.0	359.2	360.2	58.0	79.0	127.0	412.8	2	476.3	28.6	12
16"	400	595.0	35.0	36.6	457.0	410.5	411.2	64.0	87.0	127.0	469.9	2	539.8	28.6	16
18"	450	635.0	38.1	39.7	505.0	461.8	462.3	69.0	97.0	140.0	533.4	2	577.9	31.7	16
20"	500	700.0	41.3	42.9	559.0	513.1	514.4	73.0	103.0	145.0	584.2	2	635.0	31.7	20
24"	600	815.0	46.1	47.7	663.0	616.0	616.0	83.0	111.0	153.0	692.2	2	749.3	34.9	20

ASME B16.5 - Class 300 Flanges (Dimensions in mm)

NPS	DN	OD (O)	THK SO/SW (TF)	THK WN (TF)	HUB DIA (X)	BORE WN (B)	LENGTH SO/SW (Y)	LENGTH WN (Y)	RF DIA (R)	PCD (W)	HOLE (D)	NO (N)
1/2"	15	95.0	12.7	14.3	38.0	22.9	23.0	53.0	34.9	66.7	15.9	4
3/4"	20	115.0	14.3	15.9	48.0	28.2	26.0	58.0	42.9	82.6	19.0	4
1"	25	125.0	15.9	17.5	54.0	34.9	27.0	62.0	50.8	88.9	19.0	4
1 1/2"	40	155.0	19.1	20.7	70.0	50.0	31.0	69.0	73.0	114.3	22.2	4
2"	50	165.0	20.7	22.3	84.0	62.5	34.0	70.0	92.1	127.0	19.0	8
3"	80	210.0	27.0	28.6	117.0	91.4	43.0	80.0	127.0	168.3	22.2	8
4"	100	255.0	30.2	31.8	146.0	116.8	48.0	86.0	157.2	200.0	22.2	8
6"	150	320.0	35.0	36.6	206.0	171.4	53.0	99.0	215.9	269.9	22.2	12
8"	200	380.0	39.7	41.3	260.0	222.2	62.0	112.0	269.9	330.2	25.4	12
10"	250	445.0	46.1	47.7	321.0	277.4	67.0	118.0	323.8	387.4	28.6	16
12"	300	520.0	49.3	50.8	375.0	328.2	73.0	131.0	381.0	450.8	31.7	16

3. FORGED FITTINGS DIMENSIONS (ASME B16.11)

Socket Weld Fittings - Class 3000, 6000 & 9000

NOM PIPE SIZE	SOCKET BORE DIA (B)	BORE DIA FITTING (D)			SOCKET WALL THICKNESS (C)			BODY WALL THICKNESS			CENTER TO BOTTOM OF SOCKET (A)			LAYING LENGTH
		3000	6000	9000	3000	6000	9000	3000	6000	9000	ELBOW/ TEE	45° ELBOW	COUPLING (E)	HALF COUPLING (F)
1/2"	22.2	16.6	12.5	7.2	4.67	5.97	9.35	3.73	4.78	7.47	15.50	11.00	9.50	22.50
3/4"	27.6	21.7	16.3	11.8	4.90	6.96	9.78	3.91	5.56	7.82	19.00	13.00	9.50	24.00
1"	34.3	27.4	21.5	16.0	5.69	7.92	11.38	4.55	6.35	9.09	22.50	14.00	12.50	28.50
1 1/4"	43.1	35.8	30.2	23.5	6.07	7.92	12.14	4.85	6.35	9.70	27.00	17.50	12.50	30.00
1 1/2"	49.2	41.6	34.7	28.7	6.35	8.92	12.70	5.08	7.14	10.15	32.00	20.50	12.50	32.00
2"	61.7	53.3	43.6	38.9	6.93	10.92	13.84	5.54	8.74	11.07	38.00	25.50	19.00	41.00
2 1/2"	74.4	64.2	-	-	8.76	-	-	7.01	-	-	41.00	28.50	19.00	43.00
3"	90.3	79.4	-	-	9.52	-	-	7.62	-	-	57.00	32.00	19.00	44.50

Threaded Fittings - Class 2000, 3000 & 6000

NPS	CENTER TO END (A) - ELBOW/TEE			CENTER TO END (C) - CROSS			OD (H)			MIN WALL (G)		COUPLING (W)	CAP END (P)	MIN THRD LENGTH (L2)
	2000	3000	6000	2000	3000	6000	2000	3000	6000	3000	6000	3000/6000	3000/6000	MIN
1/2"	28	33	38	22	25	28	33	38	46	3.18	4.09	51	37/38	13.6
3/4"	33	38	44	25	28	33	38	46	56	3.18	4.32	51/60	41/43	13.9
1"	38	44	51	28	33	35	46	56	62	3.68	4.98	60	44/46	17.3
1 1/4"	44	51	60	33	35	43	56	62	75	3.89	5.28	67	44/48	18.4
1 1/2"	51	60	64	35	43	44	62	75	84	4.01	5.56	79	44/48	19.2
2"	60	64	83	43	44	52	75	84	102	4.27	7.14	86	48/51	19.2

4. BUTTWELD FITTINGS DIMENSIONS (ASME B16.9)

Elbows, Tees & Crosses (Center-to-End Dimensions in mm)

NPS	PIPE OD	90° LR ELBOW	45° LR ELBOW	90° SR ELBOW	90° 3D ELBOW	45° 3D ELBOW	EQUAL TEE & CROSS		180° LR RETURN	
		A	B	C	D	E	RUN (F)	OUTLET (G)	CENTER (H)	BACK-TO-FACE (J)
1/2"	21.3	38	16	-	-	25	25	25	76	56
3/4"	26.7	38	19	-	57	24	29	29	76	51
1"	33.4	38	22	25	76	31	38	38	76	56
1 1/2"	48.3	57	29	38	114	47	57	57	114	83
2"	60.3	76	35	51	152	63	64	64	152	106
2 1/2"	73.0	95	44	64	190	79	76	76	190	132
3"	88.9	114	51	76	229	95	86	86	229	159
4"	114.3	152	64	102	305	127	105	105	305	210
6"	168.3	229	95	152	457	189	143	143	457	313
8"	219.1	305	127	203	610	252	178	178	610	414
10"	273.0	381	159	254	762	316	216	216	762	518
12"	323.8	457	190	305	914	378	254	254	914	619

Concentric & Eccentric Reducers (ASME B16.9)

NOMINAL PIPE SIZE (NPS)	OUTER DIAMETER (OD)		END TO END (N)	NOMINAL PIPE SIZE (NPS)	OUTER DIAMETER (OD)		END TO END (N)
	LARGE END	SMALL END			LARGE END	SMALL END	
3/4" x 1/2"	26.7	21.3	38	4" x 3"	114.3	88.9	102
1" x 3/4"	33.4	26.7	51	4" x 2"	114.3	60.3	102
1 1/2" x 1"	48.3	33.4	64	6" x 4"	168.3	114.3	140
2" x 1 1/2"	60.3	48.3	76	8" x 6"	219.1	168.3	152
2" x 1"	60.3	33.4	76	10" x 8"	273.0	219.1	178
3" x 2"	88.9	60.3	89	12" x 10"	323.8	273.0	203

5. PIPE SCHEDULE WALL THICKNESS CHART (ANSI B36.10 / B36.19)

All dimensions are given in millimeters (mm). Wall thicknesses according to standard ASME pipe schedules.

PIPE SIZE	OD (MM)	SCH 5S	SCH 10S	SCH 20	STD	SCH 40	XS	SCH 80	SCH 120	SCH 160	XXS
1/2"	21.33	1.65	2.11	-	2.77	2.77	3.73	3.73	-	4.78	7.47
3/4"	26.67	1.65	2.11	-	2.87	2.87	3.91	3.91	-	5.56	7.82
1"	33.40	1.65	2.77	-	3.38	3.38	4.55	4.55	-	6.35	9.09
1 1/4"	42.16	1.65	2.77	-	3.56	3.56	4.85	4.85	-	6.35	9.70
1 1/2"	48.26	1.65	2.77	-	3.68	3.68	5.08	5.08	-	7.14	10.15
2"	60.33	1.65	2.77	-	3.91	3.91	5.54	5.54	-	8.74	11.07
2 1/2"	73.03	2.11	3.05	-	5.16	5.16	7.01	7.01	-	9.52	14.02
3"	88.90	2.11	3.05	-	5.49	5.49	7.62	7.62	-	11.13	15.24
4"	114.30	2.11	3.05	-	6.02	6.02	8.56	8.56	11.13	13.49	17.12
6"	168.28	2.77	3.40	-	7.11	7.11	10.97	10.97	14.27	18.26	21.95
8"	219.08	2.77	3.76	6.35	8.18	8.18	12.70	12.70	18.26	23.01	22.23
10"	273.05	3.40	4.19	6.35	9.27	9.27	12.70	15.09	21.44	28.58	25.40
12"	323.85	3.96	4.57	6.35	9.52	10.31	12.70	17.48	25.40	33.32	25.40

6. QUALITY ASSURANCE & ORDERING GUIDE

Quality Policy & Certification:

- 100% Raw material test certificates provided as per EN 10204 3.1 / 3.2.
- Hydrostatic testing, PMI testing, Ultrasonic testing, and Radiography available upon request.
- Custom machining, beveling, and specific pressure ratings executed precisely to client drawings.

How to Order / Request Quotes:

Please specify: Material Grade, Nominal Pipe Size (NPS), Class/Schedule, Standard (e.g. ASME B16.5 / B16.9), and Quantity required.